# Practice 2 Report: Foundation X+ on CANDID-PTX

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## Abstract

This report summarizes Practice 2 using Foundation X+ on CANDID-PTX for chest X-ray classification, localization, and segmentation. Seven configurations were trained: each task independently, the three pairwise combinations, and a three-task model. The purpose was to establish single-task baselines and examine whether cyclic multi-task training produces beneficial cross-task transfer. At the time of this report, the classification-only run completed 100 epochs, while the remaining runs were still in progress because of ASU Sol GPU time limits. The completed classification run achieved a best student AUC of 77.16% and best teacher AUC of 77.28%. The partial localization and segmentation baselines reached a best teacher $mAP_{40}$ of 58.27% and a best student Dice of 88.50%, respectively. Multi-task trajectories show that unfocused updates can sometimes preserve or slightly improve another task, but the current results do not yet support consistent synergy across all tasks, particularly for localization.

## 1 Introduction

Practice 2 studies Foundation X+ for learning classification, localization, and segmentation from heterogeneous annotations. The required CANDID-PTX experiments are: classification only (A), localization only (B), segmentation only (C), classification+localization (D), classification+segmentation (E), localization+segmentation (F), and classification+localization+segmentation (G).

The main goal is not only to measure task performance, but also to analyze *focused* and *unfocused* behavior. Focused performance is measured immediately after training the same task being evaluated; unfocused performance is measured after training a different task. These measurements can reveal whether cyclic training causes forgetting, interference, or useful transfer between tasks.

## 2 Hypotheses

Three behaviors were expected. First, single-task models were expected to provide strong direct baselines because every update optimizes the same objective. Second, multi-task training could temporarily reduce a task after an unfocused update, but the shared representation could also transfer useful information across classification, localization, and segmentation. Third, the teacher model was expected to be smoother than the student because it is updated through exponential moving average.

## 3 Methods

### 3.1 Dataset and Tasks

CANDID-PTX contains 19,237 chest radiographs and annotations supporting pneumothorax-related classification, localization, and segmentation. The released course split/annotation files were used. Classification performance was measured with AUC, localization with $mAP_{40}$, and segmentation with Dice score.

Table 1: Practice 2 CANDID-PTX experiments.

| ID | Tasks |
|---|---|
| A | Classification |
| B | Localization |
| C | Segmentation |
| D | Classification + Localization |
| E | Classification + Segmentation |
| F | Localization + Segmentation |
| G | Classification + Localization + Segmentation |

### 3.2 Model and Training

The released Foundation X+ implementation was used with a Swin-B backbone initialized from Ark pretrained weights. Localization used the DINO branch and segmentation used UPerNet. Images were processed at $224 \times 224$ resolution. Training used AdamW, batch size 24, 12 workers, backbone learning rate $10^{-5}$, and localization/segmentation learning rates $10^{-4}$. The student-teacher framework and Lock-Release training strategy from Foundation X+ were kept enabled.

The target horizon was 100 updates per task. Therefore, A-C used 100 task epochs, D-F used 200 cyclic epochs, and G used 300 cyclic epochs. This 100-update target was an experimental choice for a controlled comparison, not a fixed epoch requirement in the practice guide. A checkpoint was saved after each task epoch.

### 3.3 Sol Workflow

Short smoke tests were first used to verify data loading, CUDA extensions, model initialization, and each task head. Localization initially could not run because the required PNG collection was not present in the downloaded CANDID-PTX archive. The PNG images and matching localization annotations were later located in the course/lab storage and connected to the released configuration.

Long runs were submitted to ASU Sol using one A100 GPU per job. The class QOS imposed a GPU-minute limit that effectively restricted two simultaneous jobs to about 7 h 55 min each. Long experiments therefore required checkpoint/resume training. During debugging, an output-directory cleanup command also removed checkpoints before two resume attempts; this was corrected by preserving the result directory and automatically resuming from the latest fully evaluated checkpoint.

## 4 Results

### 4.1 Single-Task Baselines

Table 2 summarizes the available single-task trajectories. Only classification reached the full target of 100 updates in the extracted results.

Table 2: Single-task CANDID-PTX performance. Values are percentages.

| Run | Progress | Best Student | Best Teacher |
|---|---|---|---|
| A: CLS | 100/100 | 77.16 AUC | 77.28 AUC |
| B: LOC | 43/100 | 56.51 $mAP_{40}$ | 58.27 $mAP_{40}$ |
| C: SEG | 45/100 | 88.50 Dice | 85.97 Dice |

Classification peaked relatively early: the best student result occurred at epoch 36 and the best teacher result at epoch 33. By epoch 100, AUC had decreased to 74.07% (student) and 74.14% (teacher), indicating that additional training did not continuously improve test performance. Localization also reached its best values early in the partial run (student epoch 11, teacher epoch 16). Segmentation reached a best student Dice of 88.50% at epoch 38.

### 4.2 Multi-Task Focused and Unfocused Behavior

Table 3 shows the latest available focused and unfocused values for the student. Small differences between adjacent focused/unfocused measurements were common. For example, in D the classification AUC was 75.91% after classification and 75.99% after the adjacent localization update. In contrast, localization in D decreased from its single-task level and did not show the same benefit.

Table 3: Latest student focused (F) and unfocused (U) performance.

| Run | Task | F | U |
|---|---|---|---|
| D | CLS | 75.91 | 75.99 |
| D | LOC | 48.32 | 49.32 |
| E | CLS | 75.61 | 75.18 |
| E | SEG | 86.96 | 86.62 |
| F | LOC | 44.09 | 43.29 |
| F | SEG | 82.85 | 82.87 |
| G | CLS | 74.74 | 74.74 |
| G | LOC | 42.43 | 40.35 |
| G | SEG | 84.82 | 82.94 |

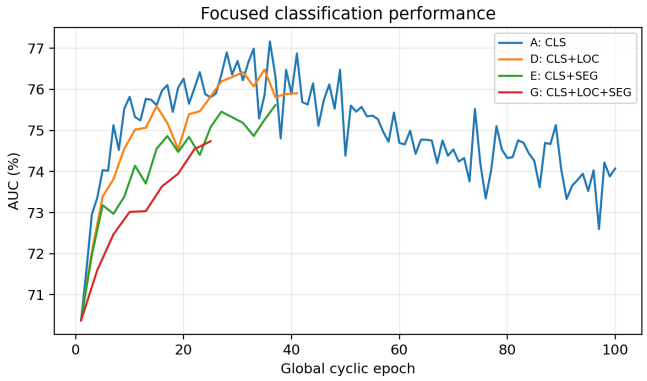



Figure 1: Focused student classification trajectories for all configurations containing classification.

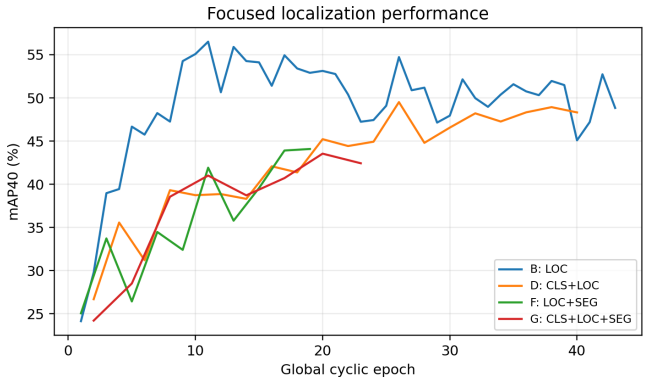



Figure 2: Focused student localization trajectories for all configurations containing localization.

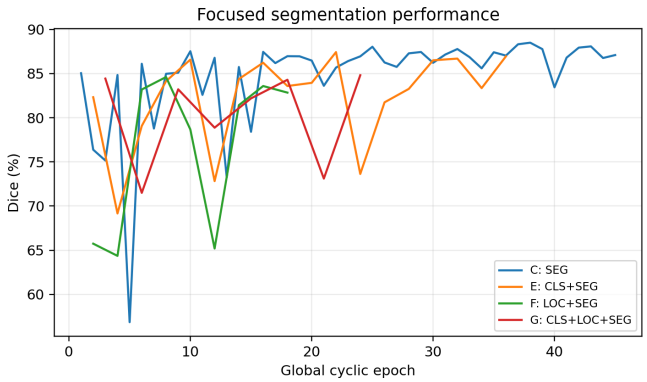



Figure 3: Focused student segmentation trajectories for all configurations containing segmentation.

### 4.3 Matched-Update Comparison

Because the multi-task models have fewer updates per task than the corresponding single-task models, comparing them only at their global epoch is not fair. Table 4 compares each latest focused *student* result with the single-task baseline after the same number of updates of that task.

Table 4: Focused multi-task student performance relative to the single-task model at the same number of task updates. $\Delta > 0$ indicates improvement.

| Run | Task | Updates | Multi | $\Delta$ |
|---|---|---|---|---|
| D | CLS | 21 | 75.91 | +0.26 |
| D | LOC | 20 | 48.32 | -4.81 |
| E | CLS | 19 | 75.61 | -0.43 |
| E | SEG | 18 | 86.96 | -0.02 |
| F | LOC | 10 | 44.09 | -11.00 |
| F | SEG | 9 | 82.85 | -2.25 |
| G | CLS | 9 | 74.74 | -0.79 |
| G | LOC | 8 | 42.43 | -4.83 |
| G | SEG | 8 | 84.82 | -0.16 |

The clearest positive student transfer in the current matched comparison is classification in D (CLS+LOC), which is 0.26 AUC points above the classification-only model after the same number of classification updates. Classification+segmentation is nearly neutral for segmentation. Localization is currently the task most negatively affected by multi-task training. These observations are preliminary because D-G are still far from the planned 100 updates per task.

### 4.4 Comparison with Published Foundation X Results

For context, the Foundation X paper reports CANDID-PTX single-task baselines of 72.61% AUC, 50.51% $\text{mAP}_{40}$, and 86.36% Dice, and Foundation X results of 73.86%, 54.14%, and 89.81%, respectively. Our best available student single-task results are 77.16% AUC, 56.51% $\text{mAP}_{40}$, and 88.50% Dice. These numbers are useful reference points but are not a strict reproduction because our training duration and exact data/configuration path differ and B/C are incomplete.

## 5 Discussion

The experiments show that the three tasks interact differently. Classification is relatively robust during multi-task training. In D, the latest classification value after localization was slightly higher than after the preceding focused classification update, and the matched-update comparison was also slightly above the classification-only baseline. This is a small example of beneficial cross-task transfer.

Segmentation is also comparatively stable. E is nearly identical to the segmentation-only baseline after the same number of segmentation updates, suggesting that adding classification does not strongly damage the segmentation representation at this stage. F and G are less mature and therefore harder to interpret.

Localization is the main weakness of the current multi-task runs. D, F, and G all trail the localization-only baseline at matched update counts. This suggests task interference, insufficient localization updates, or a need to rebalance the cyclic schedule or task-specific optimization. Therefore, the current results do not support a general claim that all three tasks synergistically improve one another. Instead, they show limited positive/neutral transfer for classification and segmentation and stronger interference for localization.

The student-teacher behavior is also task-dependent. The teacher is often smoother, but it is not always the best model. For example, the segmentation student substantially exceeds the teacher in the later single-task epochs, whereas localization shows strong teacher peaks early in training. This reinforces the importance of preserving checkpoints and selecting the best checkpoint rather than reporting only the last epoch.

## 6 Strengths, Weaknesses, and Improvements

A strength of this practice is that all seven configurations use the same dataset, backbone initialization, metrics, and optimization framework, making cross-task comparisons relatively controlled. Focused and unfocused measurements also provide more information than a single final score.

The main weakness is incomplete training for B-G. Sol wall-time and GPU-minute limits caused repeated resumes, and localization evaluation was particularly expensive. In addition, the released data paths were outdated and required manual reconstruction of the CANDID-PTX localization resources.

Possible improvements are straightforward: (1) use early stopping or best-checkpoint selection rather than assuming the final epoch is optimal; (2) rebalance task frequency or loss weights to protect localization; (3) tune task-specific learning rates; (4) add more diverse annotated datasets as done in the full Foundation X paper; and (5) compare alternative architectures such as ConvNeXt for classification, Co-DETR/Mask R-CNN for localization, and U-Net/UNet++ for segmentation.

# 7 Conclusion

Practice 2 successfully established working Foundation X+ pipelines for all seven CANDID-PTX configurations and a complete 100-epoch classification baseline. The best available single-task results were 77.16/77.28% AUC for classification (student/teacher), 56.51/58.27% $\text{mAP}_{40}$ for localization, and 88.50% student Dice for segmentation. Focused and unfocused trajectories confirm that cross-task updates do not always cause immediate forgetting, and classification in the CLS+LOC experiment shows a small positive matched-update effect. However, localization currently degrades in the multi-task settings, so the present evidence indicates a mixture of transfer and interference rather than universal synergy. Completing the remaining trajectories would be necessary for a stronger final comparison.

# References


[1] N. U. Islam, D. Ma, J. Pang, S. S. Velan, M. Gotway, and J. Liang, "Foundation X: Integrating Classification Localization and Segmentation through Lock-Release Pretraining Strategy for Chest X-ray Analysis," *Proceedings of WACV*, pp. 3647–3656, 2025.

[2] S. Feng et al., "Curation of the CANDID-PTX Dataset with Free-Text Reports," *Radiology: Artificial Intelligence*, vol. 3, no. 6, 2021.